

AI-Powered DW/DM NLP Builder - POC Generation Prompt

You are a Senior AI Architect and Python Engineer.

Build a FULLY WORKING Python Proof of Concept for an AI-Powered Data Warehouse / Data Mart NLP Builder. The POC must be runnable locally using Python. Do NOT generate pseudo-code. Generate complete source code files, project folder structure, sample data, installation instructions, README.md and requirements.txt.

Objective: Demonstrate an end-to-end AI architecture that converts business questions into SQL using metadata lookup + RAG retrieval.

Business Questions Examples: Show monthly sales by customer Top customers by revenue Product profitability analysis Revenue trend by region Customer churn analysis Sales by category
Architecture Flow:

User Question → Question Analyzer → Entity Extraction → Metric Extraction → Metadata Lookup → Relationship Resolution → RAG Retrieval → Context Builder → Prompt Builder → LLM Simulation → SQL Generation → Confidence Engine → User Approval.

Technologies: Python 3.11, SQLite, ChromaDB, Sentence Transformers, LangChain, Pandas, NumPy, scikit-learn, OpenAI-compatible abstraction layer. No cloud dependencies.

Metadata Sources: Salesforce, Epicor, Priority ERP, SYSPRO.

Required Metadata Files: semantic_entities.csv, semantic_source_mappings.csv, semantic_metrics.csv, semantic_metric_mappings.csv, semantic_relationships.csv, workspace_config.json, semantic_business_rules.txt, semantic_glossary.txt.

Implement: 1. SQLite metadata repository with deterministic lookup. 2. ChromaDB vector store. 3. Fixed, Recursive, and Metadata-aware chunking. 4. Embeddings using all-MiniLM-L6-v2. 5. Cosine similarity, Top-K retrieval, metadata filtering. 6. BM25 + Vector + Hybrid search. 7. NLP question analyzer. 8. Context builder. 9. Template-based SQL generation. 10. Confidence scoring. 11. Clarification engine. 12. End-to-end demo with logging.

Folder Structure: ai_dw_nlp_poc/ data/ metadata/ vector_store/ db/ src/ (loaders, chunking, embeddings, retrieval, metadata_lookup, analyzer, context_builder, sql_generator, confidence) run_demo.py README.md requirements.txt

Project must run with: pip install -r requirements.txt python run_demo.py

Deliver complete runnable code with no placeholders, no TODOs, and enough sample ERP metadata to demonstrate all flows.